CSCI-535 Project Final Report: Multimodal Sarcasm Detection

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1 Problem Definition

Detecting sarcasm is a challenging task, even for human interpreters, due to its reliance on subtle contextual and nonverbal cues. Several factors contribute to the difficulty of accurately identifying sarcasm:

- **Human Perception of Sarcasm:** Sarcasm is inherently ambiguous, and even people often find it difficult to understand the intended meaning behind a sarcastic remark. This ambiguity arises because the literal words may convey a different message than what is actually meant.
- **Mehrabian's Rule and Nonverbal Communication:** Mehrabian's Rule illustrates that when verbal and nonverbal signals conflict, only about 7% of the communication is carried by the words themselves, while 55% is conveyed through body language and 38% through tone of voice. This statistic underscores the critical role of nonverbal cues in understanding the true sentiment behind a statement.
- **Limitations of Single-Modality Models:** Traditional text-only models often struggle with sarcasm detection because they rely exclusively on the written word. Without access to tone, facial expressions, or body language, these models may misinterpret sarcastic remarks as sincere or neutral.
- **Advantages of Multimodal Approaches:** By incorporating multiple modalities of data, such as text, audio, and visual cues, multimodal approaches can capture a richer context and discern the nonverbal signals that are essential for accurate sarcasm detection. This integration of modalities enables a deeper understanding of the speaker's intent and emotional state.

2 Literature Review

Sarcasm detection has evolved significantly from traditional text-only approaches to sophisticated multimodal methods that incorporate visual and audio cues. This evolution reflects the complex nature of sarcasm, which often relies heavily on non-verbal signals that text-only models cannot capture.

2.1 Text-Based Sarcasm Detection

Early sarcasm detection research focused on textual cues and patterns. González-Ibánez et al. [8] pioneered work identifying patterns of hyperbole, punctuation, and sentiment shifts as indicators of sarcastic intent. Building on this, Bamman and Smith [2] demonstrated that incorporating author, audience, and contextual information improved detection accuracy over purely lexical approaches. Deep learning advancements led to more sophisticated models, with Ghosh and Veale [7] introducing a CNN-LSTM architecture capturing word-level features and sequential dependencies. Hazarika et al. [10] proposed CASCADE, a hybrid approach that uses CNN for content modeling while incorporating user embeddings and discourse features extracted from discussion forums, achieving notable accuracy improvements by capturing contextual information. Parameswaran et al. [12] further advanced this direction by applying BERT specifically to sarcasm target detection, demonstrating how transformer architectures can not only identify sarcastic content but also precisely locate the target of sarcastic remarks within text.

2.2 Visual Cues in Sarcasm Detection

Visual information provides crucial contextual clues. Schifanella et al. [14] were among the first to incorporate visual features, showing that images accompanying text could reinforce or contradict literal meanings. Their work explored multi-modal relationships on platforms like Instagram and Twitter. Recent advancements include Pan et al. [11], who proposed cross-modal attention mechanisms dynamically weighing visual features based on textual content. This approach effectively captures semantic gaps between verbal and non-verbal cues.

2.3 Audio Features in Sarcasm Recognition

Vocal tone and prosody are critical. The earliest works include Tepperman et al. [15], which identified distinctive patterns in pitch, speaking rate, and emphasis as indicators of sarcasm. Their analysis revealed sarcasm's role in dialogue grounding and acknowledgment. While Bryant [3] explored acoustic features like pitch contours and spectral analysis, recent work such as Chauhan et al. [6] emphasizes integrating prosodic information with multi-modal sentiment analysis for improved detection.

2.4 Multimodal Fusion Approaches

Multimodal integration presents challenges but offers significant improvements. Schifanella et al. [14] pioneered frameworks combining textual and visual modalities across platforms. Cai et al. [4] presented a hierarchical fusion model for Twitter data that leveraged text features, image features, and image attributes as three distinct modalities. Pan et al. [11] introduced inter-modality attention mechanisms that capture contradictions essential for sarcasm recognition. Castro et al. [5] introduced the MUStARD dataset, compiled from popular TV shows with audiovisual utterances, establishing a foundation for multi-modal sarcasm detection research. Building on this, Chauhan et al. [6] extended MUStARD with sentiment and emotion annotations, proposing two attention mechanisms that learn relationships between different segments across modalities. Various fusion strategies have emerged, from early fusion that combines raw features at the input level to late fusion where each modality is processed separately before integration. More recent approaches include complex knowledge fusion networks and transformer-based architectures that facilitate the dynamic integration of textual, visual, and audio features. These approaches emphasize modeling semantic gaps between verbal and non-verbal cues while capturing the incongruity characteristic of sarcastic expressions.

2.5 Addressing Biases and Emerging Modalities

A recent study that adds some more investigation to the field is by Qin et al. [13]. In this work, the authors identify and mitigate biases present in earlier multimodal sarcasm datasets, such as spurious correlations and imbalanced label distributions. Their refined benchmark, MMSD2.0, involves re-annotating and cleaning the original dataset to ensure that models learn genuine multimodal incongruity rather than relying on superficial cues. Additionally, the paper introduces a novel multimodal model leveraging a multi-view CLIP framework to better capture cross-modal alignment between text and images. This work not only demonstrates improved performance over previous benchmarks but also offers insights into the challenges of dataset bias and the importance of robust data curation in multimodal sarcasm detection.

3 Data Description

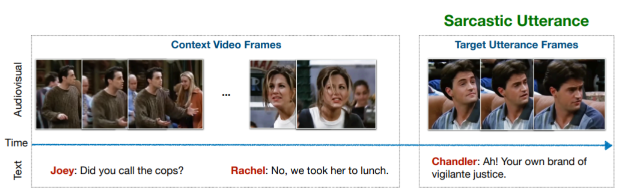

Figure 1: A sample sarcastic utterance from the MUStARD dataset showing the visual frame, textual transcript, and contextual cues.

In our work, we leverage the MUStARD (Multimodal Sarcasm Detection) dataset [5] – a novel, richly annotated resource specifically designed to capture the interplay of verbal and non-verbal cues in sarcasm. Below, we detail the key aspects of the dataset.

3.1 Data Collection and Sources

To identify potentially sarcastic utterances, the authors conducted extensive web searches using keywords such as "Friends sarcasm," "Chandler sarcasm," and "Sarcasm in TV shows." Videos were sourced from popular TV shows including *Friends*, *The Golden Girls*, *The Big Bang Theory*, and *Sarcasmaholics Anonymous*. In addition, non-sarcastic examples were obtained from MELD – a multimodal emotion recognition dataset derived from *Friends*. The initial collection comprised 6,421 videos, from which a balanced subset of 690 utterances (345 sarcastic and 345 non-sarcastic) was extracted for subsequent analysis.

3.2 Modalities and Transcriptions

Each utterance in MUStARD is composed of three distinct modalities:

- **Text:** Transcriptions of the utterances, which are obtained from available subtitles (as in the case of *The Big Bang Theory* and MELD) or via manual transcription. These transcriptions not only capture the dialogue but also preserve contextual information vital for understanding sarcasm.
- **Audio:** Speech recordings that encapsulate prosodic features (e.g., tone, pitch, and intonation). Such auditory cues help in identifying sarcastic expressions that might not be evident from the text.
- **Visual:** Video frames that provide information on facial expressions, gestures, & social context. Visual cues, such as a smirk or a straight face, serve as critical indicators of sarcastic intent.

	Utterance	Context
Unique words	1,991	3,205
Avg. length (tokens)	14	10
Max. length (tokens)	73	71
Avg. duration (s)	5.22	13.95

Table 1: Stats for utterances and their context in the MUStARD dataset.

4 Method

4.1 Extracting the Visual Embeddings

For feature extraction, we employed the Attention Mesh [9] framework part of the MediaPipe module [1], which offers robust real-time facial landmark detection. The implementation was configured to:

- Process video frames sequentially
- Detect up to one face per frame
- Extract 478 three-dimensional facial landmarks (x, y, z coordinates)
- Maintain detection and tracking confidence thresholds of 0.5
- Set `static_image_mode` to `false`, allowing for better localization of landmarks in videos.

This approach yields a comprehensive representation of facial movements and expressions throughout each utterance, capturing subtle microexpressions potentially indicative of sarcastic intent, far surpassing traditional frameworks like Dlib and OpenFace, which detect only 68 facial landmarks compared to MediaPipe's 478 3D landmarks. While Dlib struggles with accuracy during head rotations and variable lighting conditions, and OpenFace provides limited landmark resolution around crucial expressive regions like the eyes and mouth, our implementation using Attention Mesh aims for superior temporal and spatial precision for tracking the fleeting facial cues often characteristic of sarcastic delivery.

[1] https://chuoling.github.io/mediapipe/

4.1.1 Embedding Generation. The facial landmark extraction process generates a high-dimensional feature vector for each frame. Our methodology addresses the challenge of variable-length videos by implementing a frame aggregation strategy:

(1) For each video frame, we extract 478 facial landmarks with their 3D coordinates, resulting in a 1,434-dimensional vector per frame.
(2) In frames where facial detection fails, we substitute a zero vector to maintain consistent dimensionality.
(3) For the non-framewise fusion experiments involving aggregated embeddings, the frame-level embeddings are aggregated by taking the 1st PCA component across the temporal dimension, producing a single fixed-length embedding vector per video.

This aggregation strategy preserves the overall facial expression patterns while normalizing for video duration differences.

4.2 Extracting the Audio Embeddings

To complement the visual and textual modalities, we extract audio features using a two-pronged approach. First, we leverage the deep acoustic representations from a pretrained Wav2Vec2 model [1]. Both the primary *utterance* and its corresponding *context* audio files are processed through the model to obtain fixed-length 768-dimensional feature vectors for each clip.

In addition to these deep features, we compute prosodic features directly from the utterance audio to capture nonverbal cues. Specifically, we calculate the mean and standard deviation of:

- **Pitch:** Using a built-in function to detect the pitch frequency.
- **RMS Energy:** Computed over sliding windows of the audio signal via a custom function.

These handcrafted features (resulting in a 4-dimensional feature vector) help characterize the speaker's tone and energy, which are vital for discerning sarcasm. Finally, concatenating the context and audio's 772-length representation results in a 1544-dimensional feature vector per clip when we flatten along the temporal dimension. These embeddings were used to train our models in conjunction with other embeddings that were collapsed along the temporal dimension.

For framewise modeling, we extract dense temporal audio representations using a sliding window approach. Each utterance and its corresponding context clip is passed through the pretrained Wav2Vec2 model [1] to obtain deep acoustic features at 20ms resolution. These are then pooled into 100ms segments by averaging hidden states within a 240ms window sliding every 100ms, resulting in one 768-dimensional vector per frame. To enhance prosodic awareness, we append four handcrafted features—mean and standard deviation of pitch and RMS energy—computed over the same window. This yields a final per-frame representation of 772 dimensions. The utterance and context are stored as separate temporal sequences, preserving alignment for downstream frame-level fusion and cross-modal attention.

It is important to note that laugh tracks, which are prevalent in some sitcoms, may have contributed to a higher detection rate of sarcasm than expected. These laugh tracks can exhibit prosodic characteristics similar to sarcastic speech, potentially leading to false positives with a simply humorous scene being misclassified as sarcastic. It is interesting to consider that the inclusion of other modalities might serve to reduce the false positive rate and differentiate between humor and sarcasm.

4.3 Extracting the Textual Embeddings

4.3.1 ***Baseline BERT Embeddings.*** For our initial text-based experiments, we leveraged the *BERT-base* model to generate 768-dimensional embeddings. Specifically, we extracted embeddings from the `[CLS]` token by averaging its representations from the last four transformer layers. Each utterance u in the dataset, as well as its contextual text, was thus mapped to a fixed-size vector $\mathbf{u}_t \in \mathbb{R}^{768}$.

To evaluate the effectiveness of these embeddings for sarcasm detection, we employed a Support Vector Machine (SVM) with an RBF kernel, γ = "scale", and a penalty term $C = 1.0$. Training was performed using a five-fold stratified cross-validation scheme to ensure balanced distribution of sarcastic and non-sarcastic labels across folds. In each iteration, four folds were used for training (further divided into training and validation sets), while the remaining fold was used for testing. Note that this stratification process was random and did not enforce speaker-independence, meaning speakers could appear in both training and testing sets.

4.3.2 ***Multilingual Contrastively Fine-tuned Embeddings.*** Next, we investigated a more specialized embedding approach using the *Granite-Embedding-278M-Multilingual* model. This 278M-parameter model has been contrastively fine-tuned on a diverse corpus of open-source and enterprise-friendly datasets. Similar to the BERT-based method, we obtained 768-dimensional vectors per utterance by passing both the main utterance text and its context through the Granite model.

To maintain consistency and enable a fair comparison, we employed the same SVM (RBF kernel, γ = "scale", $C = 1.0$) and the same five-fold cross-validation protocol described above. This allowed us to directly contrast the performance of these multilingual, contrastively trained embeddings with those derived from BERT.

4.3.3 ***Prompt-based Classification (Gemini-1.5-Flash).*** We also explored a zero-shot classification approach using *Gemini-1.5-Flash*, a large language model accessed through a prompt-based API. Instead of generating feature embeddings for a downstream classifier, we prompted the model directly to determine whether an utterance was sarcastic. The model was given the following structured query:

> *"Analyze if the following utterance is sarcastic given the context. Context:* ***context*** *Utterance:* ***utterance*** *Answer with only 'true' if sarcastic or 'false' if not sarcastic."*

4.4 Modeling Method: Early Fusion

The Early Fusion approach combines text, audio, and visual data at the input stage. Details regarding the non-framewise and framewise implementations are given below:

4.4.1 Framewise Modeling. This variant processes sequences of audio/video frames aligned with text over time:

- Temporal alignment & Feature Combination for Modeling: We combine the temporal sequences using a bidirectional

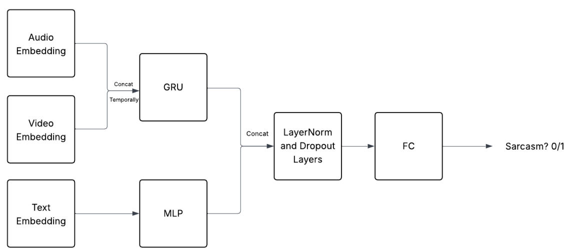

Figure 2: Modeling workflow for Framewise Early Fusion

Gated Recurrent Unit (GRU), with the audio and video features sampled at 10 FPS. The text embeddings are projected to match GRU output dimensions before concatenation. The architecture addresses variable-length sequences through padded collation.

4.4.2 Non-Framewise Modeling (on Aggregated Embeddings).

- Feature combination: Text embeddings (from BERT/Granite models), audio features (WavLM + pitch/tone metrics), and video embeddings are flattened and concatenated.
- Neural network: A multi-layer perceptron (MLP) processes concatenated feature vectors from text, audio and video (the 478 face landmarks compressed via PCA) modalities. The MLP architecture includes two hidden layers (128D and 64D) with batch normalization and dropout (p=0.5) for regularization.

4.5 Modeling Method: Late Fusion

4.5.1 Non-Framewise (on Aggregated Embeddings). Our late-fusion baseline follows a two-stage "stacked generalisation" design. Each modality is first encoded independently, producing a *256-dimensional* feature vector that is converted to a 2-logit "mini-prediction." A lightweight meta-classifier then learns to combine these four modality-specific predictions.

Per-modality encoders.

- **Visual.** A two-layer MLP (`DenseBlock`) maps the 4,784-dim face-landmark PCA vector ($2868 \to 1024 \to 256$) with ReLU and dropout (p=0.1).
- **Audio.** A shallower MLP encodes the 1,544-dim Wav2Vec2+prosody vector ($1544 \to 512 \to 256$).
- **Text (context & utterance).** A shared 1-layer MLP projects each 768-dim Granite/BERT embedding directly to 256 dims.

Stage-1 classifiers. Each 256-dim vector is mapped to 2 logits via a single linear layer $\mathrm{cls}_m : \mathbb{R}^{256} \to \mathbb{R}^2$ where $m \in \{\mathrm{vis}, \mathrm{aud}, \mathrm{ctx}, \mathrm{utt}\}$. This produces an 8-dim tensor $\mathbf{z} = [\mathbf{z}_{\mathrm{vis}}; \mathbf{z}_{\mathrm{aud}}; \mathbf{z}_{\mathrm{ctx}}; \mathbf{z}_{\mathrm{utt}}] \in \mathbb{R}^8$.

Stage-2 meta learner. A tiny MLP ($8 \to 32 \to 2$) operates on $\mathbf{z}$:

$$\mathbf{y} = f_{\mathrm{meta}}(\mathbf{z}) = \mathrm{MLP}_{8\to32\to2}(\mathbf{z}),$$

where $\mathbf{y}$ are the final sarcasm logits.

Modality dropout. During training, we randomly mask each $\mathbf{z}_m$ with probability p_{drop}=0.1 ("logit dropout"). This regularises the

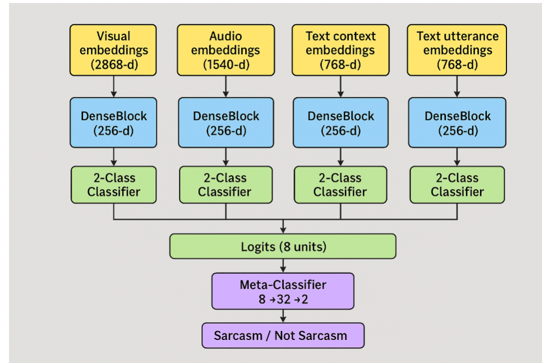

Figure 3: Modeling workflow for Non-Framewise Late Fusion

meta learner and encourages robustness to missing modalities without detaching gradients.

Optimisation details. We train for 20 epochs with AdamW (η=3×10^{-4}, batch size 64) and cross-entropy loss. The best model on the validation set (macro F1) is checkpointed and later evaluated on the withheld test split.

4.5.2 Framewise. For the framewise scenario, we decided to adopt for a *hard voting ensemble*. We run three independent models and use thier outputs as the votes:

(1) The **audio GRU** from Section 4.5.1 (framewise early fusion),
(2) The **video GRU** from Section 4.5.1, and
(3) The **non-framewise late-fusion** model described above

For each sample, we collect the binary predictions of the three models and return the majority class. Ties are impossible as we have an odd number of votes. Because the models are complementary (audio handles tone, video catches facial expressions, and the combined model works well across all types), this simple approach gives strong results without needing any extra training.

Why hard voting? We wanted to directly use the outputs of the existing framewise models, as they are already complex and well-engineered. Instead of designing a new trainable meta-learner—which quickly overfit the small set of 345 sarcastic examples in MUStARD—we opted for hard voting. This approach requires *no* additional parameters and still gave a consistent gain of +1.2,pp in macro F1 over any individual framewise model.

4.6 Modeling Method: Attention Fusion

The Attention Fusion paradigm emphasizes learning inter-modal relationships via attention mechanisms rather than simple concatenation. We present both framewise and non-framewise implementations below:

4.6.1 Framewise Modeling. This approach models time-aligned sequences using attention across modalities:

- Frame Projection and Encoding: Audio and video features sampled at 10 FPS are concatenated per frame and projected via a shared linear layer. These are split into utterance and context segments and encoded using bidirectional GRUs.

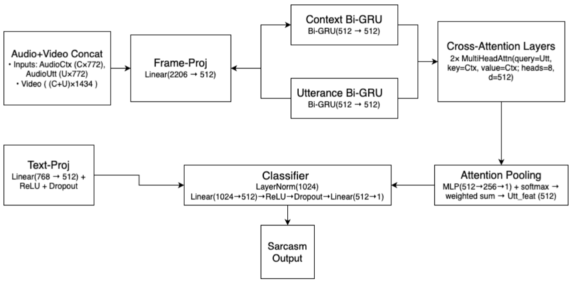

Figure 4: Modeling workflow for Framewise Cross Attention

- Cross-Attention Mechanism: The utterance GRU output attends to the context via stacked multi-head attention layers (8 heads, 2 layers), learning cross-modal and contextual dependencies.
- Attention Pooling and Fusion: A learnable scalar attention mechanism computes a weighted summary of the utterance frames. This pooled vector is concatenated with a projected text embedding and passed through a final MLP for classification.

We later ablate each of these components (Section 3) to quantify their individual impact.

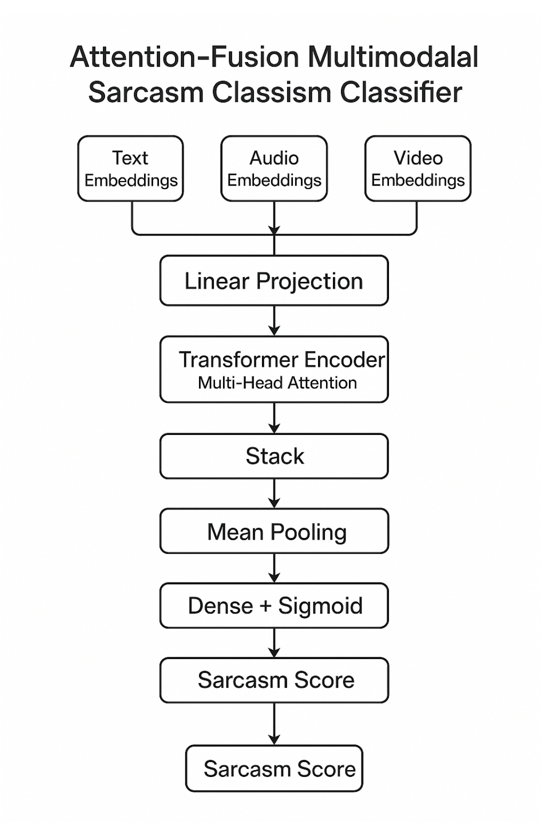

Figure 5: Modeling workflow for Aggregated Attention Fusion

4.6.2 *Non-Framewise Modeling (on Aggregated Embeddings).*

- Modality Projection: Text, audio, and video embeddings are projected into a shared 128-dimensional space via layers with ReLU, batch normalization, and dropout
- Transformer Fusion: Three modality embeddings are treated as tokens and passed through 2-layer Transformer (4 heads)
- Pooling and Classification: Outputs are mean-pooled and fed into an MLP (128 → 64 → 1) for binary sarcasm prediction.

5 Results

Method	Acc.	Prec.	Recall	F1
Early Fusion	**0.7101**	**0.667**	**0.8571**	**0.7590**
Late Fusion Ensemble	0.6812	0.6857	0.6857	0.6857
Attention Fusion (not framewise)	0.710	0.692	0.771	0.730
Early Fusion (Framewise)	0.6667	0.6579	0.7143	0.6849
Late Fusion Voting (Framewise)	0.594	0.569	0.829	0.674
Attention Fusion (Framewise)	0.696	0.667	0.800	0.727
Gemini 2.0 Flash (for comparison)	0.6812	0.6275	0.9143	0.7442
Gemini 2.0 Pro (for comparison)	0.7971	0.7143	1.000	0.893
MUStARD SVM (baseline)	N/A	0.643	0.626	0.628

Table 2: Comparison of methods on evaluation metrics.

The results table highlights how different fusion strategies performed on the sarcasm detection task. Early fusion (non-framewise) stood out with an F1 score of 0.75, showing that combining text, audio, and visual features early on helps the model pick up on the subtle cues that signal sarcasm. This approach worked especially well because it allowed the model to learn from the interactions between different types of data, which is important in a dataset like MUStARD where context is often brief and nuanced.

On the other hand, the framewise early fusion method, which tries to follow changes in audio and video over time, didn't perform quite as well (0.6849 F1 Score). This is likely because sequence models like GRUs need more data to really understand longer patterns, and our dataset is fairly small. Late fusion methods, which combine predictions from separate models for each modality, also lagged behind, probably because they miss out on the deeper connections between modalities. Notably, a simpler mean-pooling variant of the frame-wise attention model outperformed the full version, a finding we explore in the ablation study below.

Interestingly, attention-based fusion methods and large vision-language models like Gemini Pro performed strongly, but they come with higher computational costs. Overall, the results show that early fusion offers a good balance between accuracy and efficiency for sarcasm detection, especially when working with limited data and resources.

5.1 Ablation Study

Table 3 shown below demonstrates that running an ablation study as recommended during our presentation on the framewise cross attention model does result in improvements and tips it into being our best model. The study was done by taking the architecture discussed in our Methods section under 4.6.1 and retraining for 30 epochs, either with GRUs removed or replacing the learnable

Variant	Acc.	Prec.	Rec.	F1
Early Fusion (best pre-ablation)	0.7101	0.667	0.8571	0.7590
Full model (Sec. 4.6.1)	0.696	0.640	0.914	0.753
No Context GRU	0.652	0.608	0.886	0.721
No Utterance GRU	0.681	0.659	0.771	0.711
Mean Pool	**0.725**	**0.667**	**0.914**	**0.771**
No Ctx/Uttr GRU	0.609	0.567	0.971	0.716
No Ctx/Uttr GRU w Mean Pool	0.638	0.596	0.886	0.713

Table 3: Frame-wise cross-attention ablations.

attention pooling layer with a simpler mean pool. All variants tested and corresponding metrics are shown.

Key takeaways.

- Replacing the learnable attention pool with a simple mean pool slightly *improved* F1 (+1.8pp) and accuracy (+3pp), indicating the model may have over-fitted the small training set when given extra parameters.
- The context Bi-GRU contributes more than the utterance Bi-GRU: dropping it alone hurts F1 by 3pp. This is in line with intution given that the utterance clip is usually much shorter than the context clip.
- Eliminating *both* GRUs but keeping mean pooling recovers some performance, suggesting that temporal encoding and pooling provide complementary gains.

6 Conclusions

This project demonstrates the value of integrating multiple modalities-text, audio, and video cues for detecting sarcasm in conversational data. Our experiments show that fusion methods, particularly the non-framewise approach, are effective at leveraging the complementary strengths of each modality, resulting in good accuracy and F1 scores. The non-framewise early fusion model was able to capture the subtle interplay between spoken words, tone, and facial expressions, which is crucial for identifying sarcasm, especially in short and context-dependent exchanges.

While framewise early fusion and attention-based models offer the potential to model temporal dynamics and cross-modal relationships more deeply, their performance was limited by the relatively small size of the MUStARD dataset and the challenges of learning long-range dependencies with sequence models. Our ablation shows that, on MUStARD's limited data, removing learnable pooling parameters actually helps generalization—a reminder that simpler inductive biases can outperform heavier architectures when data are scarce.

Overall, this work supports the growing consensus in recent research that multimodal approaches are necessary for reliable sarcasm detection, as they better reflect the complex, real-world ways in which sarcasm is communicated.

7 Reproducibility

Our full code and data-processing scripts are publicly available at `https://github.com/anupampatil44/CSCI-535-Project`.

8 Team Members' Contributions

- **Abi Murthy**: Worked on the audio modality implementation & Attention Fusion, including the extraction of WavLM embeddings, prosodic feature engineering, and audio classifier development. Conducted ablation study.
- **Anupam Patil**: Worked on visual data processing & Early Fusion, their embedding aggregation strategies, and visual feature classification. (Refer to midterm report)
- **Nick Prestine**: Worked on textual modality & Late Fusion, implementing baseline BERT embeddings with SVM, contrastively fine-tuned Granite-Embedding with SVM, and zero-shot classification using Gemini-1.5-Flash. (Refer to midterm report)

All have contributed equally to the report writing.

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